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Publications & Conferences

Investigating soil-plant-atmosphere interactions by combining spectral electrical impedance tomography with environmental and physiological timeseries (conference, invited) Feb. 2026

Valentin Michels, Simon De Cannière, Gina Lopez, Maximilian Weigand, Kevin Warstat, Sabine Seidel, Onno Muller, Uwe Rascher, Harry Vereecken, Andreas Kemna

Agrogeophysics Workshop, Lisbon, Portugal

Non-invasive phenotyping of sugar beet and maize roots using field-scale spectral electrical impedance tomography (article) Jul. 2025

Valentin Michels, Maximilian Weigand, Lena Lärm, Onno Muller, Andreas Kemna

Plant, Cell & Environment – 10.1111/pce.70049

From root to leaf: Multi-sensor monitoring of the soil-plant-atmosphere continuum (conference, invited) Apr. 2025

Valentin Michels, Simon De Cannière, Gina Lopez, Maximilian Weigand, Kevin Warstat, Bastian Siegmann, Sabine Seidel, Onno Muller, Uwe Rascher, Harry Vereecken, Andreas Kemna

EGU General Assembly, Vienna, Austria – 10.5194/egusphere-egu25-8723

Quantitative phenotyping of crop roots with spectral electrical impedance tomography: a rhizotron study with optimized measurement design (article) Aug. 2024

Valentin Michels, Chunwei Chou, Maximilian Weigand, Yuxin Wu, Andreas Kemna

Plant Methods – 10.1186/s13007-024-01247-7

Monitoring and modelling the diurnal dance between photosynthesis and transpiration (conference) May 2024

Simon De Canniere, Bastian Siegmann, Valentin Michels, Gina Lopez, Michael Herbst, Andreas Kemna, Harry Vereecken, François Jonard, Sabine Seidel

4th VOD 4 Forest Workshop, Oberpfaffenhofen, Germany

Investigating electrical polarization signatures of sugar beet and maize: A field study using spectral electrical impedance tomography (conference) Mar. 2024

Valentin Michels, Maximilian Weigand, Andreas Kemna

EGU General Assembly, Vienna, Austria – 10.5194/egusphere-egu24-16788

Quantitative phenotyping of plant roots using spectral electrical impedance tomography (conference) Feb. 2024

Valentin Michels, Chunwei Chou, Maximilian Weigand, Yuxin Wu, Andreas Kemna

Agrogeophysics Workshop, Zürich, Switzerland – 10.62329/EXVV3683

Rhizotron Scale Imaging of Crop Root Systems Using Spectral Electrical Impedance Tomography With Optimized Measurement Configurations (conference) Dec. 2023

Valentin Michels, Chunwei Chou, Maximilian Weigand, Yuxin Wu, Andreas Kemna

AGU Fall Meeting, San Francisco, United States

Design and operation of a long-term monitoring system for spectral electrical impedance tomography (sEIT) (article) Nov. 2022

Maximilian Weigand, Egon Zimmermann, Valentin Michels, Johan Alexander Huismann, Andreas Kemna

Geosci. Instrum. Method. Data Syst. – 10.5194/gi-11-413-2022

Sensing of Winterwheat Root-systems at the Field Scale Using Spectral Electrical Impedance Tomography (conference) Mar. 2022

Valentin Michels, Maximilian Weigand, Andreas Kemna

Agrogeophysics Workshop, Brussels, Belgium